Twisted-Convolution Identities in Dimensions Seven and Eight: Shintani Height Rigidity and CM Descent

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Abstract

We prove the formal Twisted Convolution Conjecture of Appleby–Flammia–Kopp unconditionally for the discriminant-32 dimension-seven stratum and for every admissible tuple in dimension eight. In dimension seven, conductor lowering places every characteristic in one of six maximal-order ray groups. Shintani's proved index-two algebraicity theorem gives a safe exponent 16128; rigorous Arb enclosures, Voutier's height gap, exact Artin labels, and a degree-48 compositum certificate identify the complete signed overlap packet and make all 441 rank-two minors vanish for both formal shifts. Dimension eight has two admissible form conductors. For conductor three, genuine linear reinduction to quartic ray characters over $\mathbb{Q}(\sqrt{-6})$ and Stark's proved imaginary-quadratic rank-one theorem determine the two formerly missing orientations. For conductor one, the relevant ray characters are quadratic; analytic class-number formulas determine their units, and an exact phase calculation in $\mathbb{Q}(\sqrt{5})$ orients all 63 nonzero cocycle values. Exact quotient-ring arithmetic makes all 784 minors vanish for both shifts in each dimension-eight stratum. No unproved Stark conjecture or numerical recognition is used as a logical input. Dimension-seven admissibility also permits discriminant 8; that second stratum is isolated explicitly but is not claimed here.

1 Formal TCC and conventions

We use the conventions of Appleby–Flammia–Kopp (AFK) [1]. An admissible tuple $t = (d, r, Q)$ contains a primitive indefinite integral binary quadratic form Q, a positive real fixed point ρ_t, a level-d stabilizer A_t, and the matrix $L_{z,t}$. For $\boldsymbol{p} \in \mathbb{Z}^2$, let $\mathcal{I}_{\boldsymbol{p}}$ be a complete set of representatives modulo d containing $\mathbf{0}$ and $\boldsymbol{p}$. A residue $\lambda \in \mathbb{Z}/d\mathbb{Z}$ is a *shift* when its AFK coprimality condition holds and

$$\sum_{\boldsymbol{q}\in\mathcal{I}_{\boldsymbol{p}}} \omega_d^{r\langle \boldsymbol{p},(\lambda I + L_{z,t})\boldsymbol{q}\rangle} \operatorname{shin}_{A_t}^{\boldsymbol{q}/d}(\rho_t) \operatorname{shin}_{A_t^{-1}}^{(\boldsymbol{q}-\boldsymbol{p})/d}(\rho_t) = d^2 \delta_{\boldsymbol{p},\mathbf{0}}^{(d)} \tag{TCC}$$

for every $\boldsymbol{p}$. Write $\mathcal{Z}_t$ for the shift set. The formal TCC asserts $0, 1 \in \mathcal{Z}_t$ and equality of shift sets for admissible forms of the same discriminant.

Our reciprocal double sine is

$$S_2(z \mid \omega_1, \omega_2) := \frac{\Gamma_2(z \mid \omega_1, \omega_2)}{\Gamma_2(\omega_1 + \omega_2 - z \mid \omega_1, \omega_2)}. \tag{1}$$

It is the reciprocal of Kopp's $\mathrm{Sin}_{2,\mathrm{Kopp}}$ convention [2, Definition 4.22]. In particular,

$$S_2(z + 1 \mid \beta, 1) = 2\sin(\pi z/\beta) S_2(z \mid \beta, 1). \tag{2}$$

The place labels ∞_1, ∞_2 always refer to the positive and negative square-root embeddings printed in the certificates. PARI orders the negative root first in the quadratic fields used below; all raw PARI conductor vectors are translated to the displayed place labels before use.

Proposition 1 (Finite reconstruction bridge)**.** *Suppose the convention-matched AFK values $\nu_{\boldsymbol{p}}$ are inserted into the normalized Weyl reconstruction for a permitted twist. Then the reconstructed matrix* Π *satisfies* $\operatorname{Tr}\Pi = 1$*, and equation* (TCC) *is equivalent to* $\Pi^2 = \Pi$*. If all rank-two minors vanish and* $\Pi \neq 0$*, then* Π *is a rank-one idempotent and the corresponding residue is a shift.*

Proof. Expand the Weyl coefficient of $\Pi^2 - \Pi$, use the Weyl multiplication law, and reindex by the displacement difference. The AFK endpoint correction changes the raw diagonal value $\sqrt{d+1}$ to the normalized zero overlap one. The resulting coefficient is exactly the left side of (TCC) minus its $d^2\delta$ term. Fourier inversion is injective. Finally, vanishing of the 2×2 minors gives rank at most one, while trace one gives rank exactly one and idempotency. □

Theorem 2. *For every admissible dimension-seven tuple t whose form has discriminant* 32*, and for every admissible dimension-eight tuple t,*

$$0, 1 \in \mathcal{Z}_t.$$

If t, t' have forms of the same discriminant, then $\mathcal{Z}_t = \mathcal{Z}_{t'}$.

2 Dimension seven

2.1 Admissible strata and scope

The admissibility recurrence forces

$$d = 7 \Longrightarrow r = 1, \quad n = 8, \quad K_7 = \mathbb{Q}(\sqrt{2}), \quad j = m = 1, \quad f_1 = 2.$$

The form conductor may therefore be $f = 1$ or $f = 2$. Both strata exist; convenient representatives are

$$Q_{7,1} = \langle 1, -4, 2\rangle, \quad \operatorname{disc} Q_{7,1} = 8, \qquad Q_{7,2} = \langle 1, -6, 1\rangle, \quad \operatorname{disc} Q_{7,2} = 32.$$

This section proves the formal TCC for $Q_{7,2}$ and transports it within discriminant 32. It does not use form covariance to cross from discriminant 32 to discriminant 8, and it makes no claim for $Q_{7,1}$.

2.2 Conductor lowering and ray classes

Put

$$K_7 = \mathbb{Q}(\sqrt{2}), \qquad \alpha = 2 + \sqrt{2}, \qquad \beta_7 = 3 + 2\sqrt{2}.$$

Here β_7 is the positive fixed point of $Q_{7,2}$, whereas α is the reduced maximal-order point used only after conductor lowering. With

$$L_7 = \begin{pmatrix} 6 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A_{7,2} = L_7^3 = \begin{pmatrix} 204 & -35 \\ 35 & -6 \end{pmatrix},$$

the exact Rademacher calculation gives

$$\Phi_{\boldsymbol{p}} = \zeta_{56}^{7 - 32 Q_{7,2}(\boldsymbol{p})}.$$

The determinant-two map

$$B = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}$$

sends α to β_7. For a characteristic $(a,b)/7$, its two preimages are

$$s_j = \frac{1}{14}\begin{pmatrix} a+b+7j \\ 2b \end{pmatrix}, \qquad j = 0, 1. \tag{3}$$

Kopp's inverse ray map assigns to s_j the principal element

$$\gamma_{a,b,j} = 2b\sqrt{2} + 3b - a - 7j. \tag{4}$$

Removing the common ideal divisor of (14) and (γ) produces exactly six lowered moduli. Their one-place ray groups and Kopp exponents are

modulus HNF	ray group	n
$\left(\begin{smallmatrix} 14 & 0 \\ 0 & 14 \end{smallmatrix}\right)$	$C_6 \times C_2$	1
$\left(\begin{smallmatrix} 7 & 0 \\ 0 & 7 \end{smallmatrix}\right)$	C_6	1
$\left(\begin{smallmatrix} 14 & 6 \\ 0 & 2 \end{smallmatrix}\right)$	C_2	1
$\left(\begin{smallmatrix} 14 & 8 \\ 0 & 2 \end{smallmatrix}\right)$	C_2	1
$\left(\begin{smallmatrix} 7 & 3 \\ 0 & 1 \end{smallmatrix}\right)$	C_2	2
$\left(\begin{smallmatrix} 7 & 4 \\ 0 & 1 \end{smallmatrix}\right)$	1	2.

(5)

The common determinant-one stabilizer of the lowered factors at α is

$$A_{\mathrm{low}} = \begin{pmatrix} 239 & -140 \\ 70 & -41 \end{pmatrix}. \tag{6}$$

The exact inverse-ray, gcd, sign-class, and stabilizer computations are printed for all 96 lifted factors by `scripts/explore_dimension_seven_conductor_lowering.gp`.

2.3 Powered algebraicity and packet identification

The full one-place ray-14 field has relative group $C_6 \times C_2$. In its normal closure, the maximal absolutely abelian subfield has index two and the applicable imaginary-quadratic base is $\mathbb{Q}(\sqrt{-7})$. Auditing all 32 divisors of its conductor gives

$$m_{\mathfrak{d}} = \begin{cases} 12h_{\mathbb{Q}(\sqrt{-7})}n(S_{\mathfrak{d}}), & \mathfrak{d} = (1), \\ 12f_{\mathfrak{d}}n(S_{\mathfrak{d}}), & \mathfrak{d} \neq (1), \end{cases} \qquad m = \mathrm{lcm}_{\mathfrak{d}|\mathfrak{c}}\, m_{\mathfrak{d}} = 16128. \tag{7}$$

Write $\mathfrak{c} = \mathfrak{p}_2^3\bar{\mathfrak{p}}_2^3\mathfrak{p}_7$ and $\mathfrak{d}(e_1, e_2, e_7) = \mathfrak{p}_2^{e_1}\bar{\mathfrak{p}}_2^{e_2}\mathfrak{p}_7^{e_7}$. The complete divisor audit is below. Each triple records $(|H_{\mathfrak{d}}|, n(S_{\mathfrak{d}}), m_{\mathfrak{d}})$; here $|H_{\mathfrak{d}}|$ is the ray-group order, and w_0/w_1 records $w(\mathfrak{d})$ for $e_7 = 0/1$. The two triple columns have $e_7 = 0$ and $e_7 = 1$, respectively.

e_1	e_2	w_0/w_1	$e_7 = 0$	$e_7 = 1$
0	0	2/1	$(1, 96, 1152)$	$(3, 16, 1344)$
0	1	2/1	$(1, 96, 2304)$	$(3, 16, 2688)$
0	2	1/1	$(1, 48, 2304)$	$(6, 8, 2688)$
0	3	1/1	$(2, 24, 2304)$	$(12, 4, 2688)$
1	0	2/1	$(1, 96, 2304)$	$(3, 16, 2688)$
1	1	2/1	$(1, 96, 2304)$	$(3, 16, 2688)$
1	2	1/1	$(1, 48, 2304)$	$(6, 8, 2688)$
1	3	1/1	$(2, 24, 2304)$	$(12, 4, 2688)$
2	0	1/1	$(1, 48, 2304)$	$(6, 8, 2688)$
2	1	1/1	$(1, 48, 2304)$	$(6, 8, 2688)$
2	2	1/1	$(2, 24, 1152)$	$(12, 4, 1344)$
2	3	1/1	$(4, 12, 1152)$	$(24, 2, 1344)$
3	0	1/1	$(2, 24, 2304)$	$(12, 4, 2688)$
3	1	1/1	$(2, 24, 2304)$	$(12, 4, 2688)$
3	2	1/1	$(4, 12, 1152)$	$(24, 2, 1344)$
3	3	1/1	$(8, 6, 576)$	$(48, 1, 672)$

Thus the lcm of all thirty-two displayed exponents is 16128. The full HNF ideals and independently recomputed ray orders are in the archived transcript

`certificates/dimension-seven-shintani-divisors.txt`.

Here the unit-congruence factor $w(\mathfrak{d})$ is computed separately for every divisor; four divisors, including the trivial one, have $w = 2$. Shintani's proved theorem [4] therefore makes the powered packet algebraic.

The sixteen squared-overlap representatives satisfy the reciprocal polynomial

$$\begin{aligned} P(X) = X^{16} &- (37 + 28\sqrt{2})X^{15} + (1212 + 854\sqrt{2})X^{14} \\ &- (20685 + 14630\sqrt{2})X^{13} + (210371 + 148750\sqrt{2})X^{12} \\ &- (1313872 + 929054\sqrt{2})X^{11} + (5031845 + 3558044\sqrt{2})X^{10} \\ &- (11531249 + 8153832\sqrt{2})X^{9} + (15288774 + 10810788\sqrt{2})X^{8} \\ &- (11531249 + 8153832\sqrt{2})X^{7} + (5031845 + 3558044\sqrt{2})X^{6} \\ &- (1313872 + 929054\sqrt{2})X^{5} + (210371 + 148750\sqrt{2})X^{4} \\ &- (20685 + 14630\sqrt{2})X^{3} + (1212 + 854\sqrt{2})X^{2} \\ &- (37 + 28\sqrt{2})X + 1. \end{aligned}$$

It factors over $\mathbb{Q}(\sqrt{2})$ with degrees $2, 2, 12$. Exact class-field comparisons identify those factors with the two quadratic lowered fields P_+, P_- and the one-place ray-14 field. The following rational intervals each contain exactly one real root by Sturm's theorem; the last column is the exact ray log or quadratic class label.

(a,b)	interval	label	(a,b)	interval	label
$(2,2)$	$(.043513, .043514)$	$[2,1]$	$(1,4)$	$(.079508, .079509)$	$P_+ : [1,1]$
$(1,3)$	$(.080124, .080125)$	$[3,0]$	$(1,2)$	$(.104688, .104689)$	$P_- : [1]$
$(1,1)$	$(.146404, .146405)$	$[4,1]$	$(0,6)$	$(.169377, .169378)$	$[3,1]$
$(0,5)$	$(.315227, .315228)$	$[1,1]$	$(0,4)$	$(.671533, .671534)$	$[2,0]$
$(0,3)$	$(1.489129, 1.489130)$	$[5,1]$	$(0,2)$	$(3.172313, 3.172314)$	$[4,0]$
$(0,1)$	$(5.903986, 5.903987)$	$[0,0]$	$(2,6)$	$(6.830385, 6.830386)$	$[1,0]$
$(3,6)$	$(9.552165, 9.552166)$	$P_- : [0]$	$(4,4)$	$(12.480646, 12.480647)$	$[0,1]$
$(3,5)$	$(12.577346, 12.577347)$	$P_+ : [0,0]$	$(4,5)$	$(22.981630, 22.981631)$	$[5,0]$

The split primes above 17 and 41 have ray logs $[1,0]$ and $[0,1]$ and act by automorphisms 5 and 10, respectively. Local Frobenius congruences certify these actions and every table label in `scripts/dimension_seven_artin_labels.gp`.

Rigorous Arb evaluation of the eight independent analytic values gives the raw logarithmic enclosure

$$\epsilon_{\log} \le 5.86 \cdot 10^{-11}, \qquad 16128\epsilon_{\log} \le 9.45 \cdot 10^{-7}.$$

The complete balls are archived in

`certificates/dimension-seven-double-sine-intervals.txt`.

Voutier's lower bound [5] is greater than $5.22 \cdot 10^{-5}$ for every relevant degree. The quotient of an analytic value by its assigned algebraic root must therefore be a root of unity; positivity and the exact cocycle multipliers select the displayed root.

2.4 Exact finite certificate

Let H be the degree-24 signed overlap field and $C = \mathbb{Q}(\zeta_{56})$. Their intersection has degree 12. The standard component is selected by the exact identity

$$c_H = \zeta_{56} + \zeta_{56}^{-1}, \qquad 1.98 < c_H < 1.99, \tag{8}$$

and it also satisfies

$$\sqrt{2} = \zeta_{56}^{7} + \zeta_{56}^{-7}. \tag{9}$$

In the resulting degree-48 compositum, `scripts/dimension_seven_exact_tcc.gp` verifies for each formal shift

$$\operatorname{Tr}\Pi = 1, \qquad \Pi^2 = \Pi, \qquad 441 \text{ rank-two minors vanish.} \tag{10}$$

Proposition 1 proves both shifts for the discriminant-32 tuple $Q_{7,2}$. The order of discriminant 32 has wide class number one. Unconditional $\mathrm{GL}_2(\mathbb{Z})$ covariance therefore transports the result to every admissible dimension-seven form of discriminant 32, and no farther. The discriminant-8 stratum requires an independent analytic packet and remains open.

3 Dimension eight

Dimension-eight admissibility has $j = 2$ and $f_2 = 3$. The form conductor is therefore one or three, giving discriminants 5 and 45. These strata must be proved independently.

3.1 Conductor three: linear CM reinduction

For the canonical discriminant-45 form, the order-ray group is identified with the maximal-order ray group

$$\mathrm{Cl}_{(8)\infty_2}(\mathbb{Z} + 3\mathcal{O}_{\mathbb{Q}(\sqrt{5})}) \simeq \mathrm{Cl}_{(24)\infty_2}(\mathcal{O}_{\mathbb{Q}(\sqrt{5})}) \simeq C_4 \times C_2^2. \tag{11}$$

Kopp's conductor-lowering formula supplies the exact three-factor dictionary. The fifteen lower characteristics lie in an index-two ray-12 field; Shintani exponent 576, Arb isolation, and height rigidity prove that entire lower packet.

The primitive packet contains two conjugate pairs of quartic characters. On the full two-place ray group, base conjugation does not literally send either one-place character to its inverse because it swaps the labeled infinite place. The projective quotient is nevertheless the same order-two character in both cases, and the associated biquadratic field has quadratic subfields

$$\mathbb{Q}(\sqrt{5}), \qquad \mathbb{Q}(\sqrt{-6}), \qquad \mathbb{Q}(\sqrt{-30}). \tag{12}$$

This projective observation is not used as a substitute for linear equivalence. The script `scripts/dimension_eight_linear_cm_reinduction.gp` compares the complete induced characters on the degree-16 normal closure and proves genuine linear reinduction, without a scalar twist, from quartic ray characters over $\mathbb{Q}(\sqrt{-6})$. The alternative base $\mathbb{Q}(\sqrt{-30})$ gives an independent check.

Lemma 3 (Ray labels and local factors)**.** *Let*

$$\mathfrak{m}_M = \mathfrak{p}_2^3 \mathfrak{p}_3 \mathfrak{p}_5,$$

with no infinite component, where the prime ideals are the exact factors printed by `scripts/dimension_eight_cm_unit_lattice.gp`*. The ray group* $\mathrm{Cl}_{\mathfrak{m}_M}(M)$ *has cyclic coordinates* $[8, 4]$*, and in those coordinates the reinduced characters are*

$$\psi_0 = [6, 1], \qquad \psi_1 = [2, 1]. \tag{13}$$

They have the same complete Artin L-functions as the original $\mathbb{Q}(\sqrt{5})$ *characters* $[1,0,0]$ *and* $[1,1,0]$, *respectively. With*

$$S_M = \{\infty, \mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}_5\}, \tag{14}$$

every finite member is ramified for ψ_b, *so* $L_{M,S_M}(s,\psi_b) = L_M(s,\psi_b)$.

Proof. The exact normal-closure calculation leaves precisely two faithful ray characters for each kernel, an inverse pair. For packet zero the coefficient at $n=5$ is $-i$ for $[6,1]$ and $+i$ for $[2,3]$; the source coefficient is $-i$. For packet one it is $+i$ for $[2,1]$ and $-i$ for $[6,3]$; the source coefficient is $+i$. This single exact separating coefficient selects among an exhaustive two-element set; it is not finite-coefficient recognition. `scripts/dimension_eight_cm_unit_lattice.gp` fixes the relative factors before constructing their ray groups and prints these four coefficients.

The two source characters have full conductor $(24)\infty_2$. The same audit proves that both CM conductors equal $\mathfrak{m}_M$ and that ψ_b is ramified at every finite member of S_M. Thus those local factors are already one. Linear reinduction gives the equality of the complete functions $L_M(s,\psi_b) = L_K(s,\chi_b)$; no unramified source Euler factor at 5 is removed. The archimedean parity is part of the same induced character equality. □

Lemma 4 (Normalized imaginary-quadratic Stark resolvent)**.** *Let* $M = \mathbb{Q}(\sqrt{-6})$ *and let* E_b/M *be either cyclic quartic extension selected in Lemma 3. There is a global unit* $\varepsilon_b \in E_b^\times$ *such that, for* $g \in \mathrm{Gal}(E_b/M)$,

$$\zeta'_{S_M}(0,g) = -\log|g(\varepsilon_b)|_{\mathrm{ord}}, \qquad |z|_{\mathrm{ord}} := \sqrt{z\bar{z}}. \tag{15}$$

Consequently

$$L'_{S_M}(0,\psi_b) = -\sum_g \overline{\psi_b(g)} \log|g(\varepsilon_b)|_{\mathrm{ord}}. \tag{16}$$

Proof. Stark's proved rank-one theorem for abelian extensions of an imaginary quadratic base [6] applies to E_b/M with the set (14). Here $|S_M| = 4 \geq 3$, so the unit clause in Stark's theorem gives a global unit rather than only an S-unit. The script `scripts/dimension_eight_cm_unit_lattice.gp` independently certifies `bnfcertify=1` and $e = |\mu(E_b)| = 2$ for both fields. Stark's normalized complex-place absolute value is $|z|^2_{\mathrm{ord}}$; hence its factor $1/e$ becomes coefficient one in the ordinary modulus convention used above. The finite set contains the distinct conductor primes above $2,3,5$. Fourier transforming the displayed partial-zeta formula gives (16). □

Exact unit-action matrices show that the quartic anti-unit component in Lemma 4 is a rank-two integral lattice. Rigorous Arb inversion isolates the four coordinate pairs

$$(0,2), \quad (-2,0), \quad (0,-2), \quad (2,0) \tag{17}$$

with radii below $5 \cdot 10^{-9}$. Exact identities in the common normal closure then match their absolute norms to the original real-quadratic units. Thus the two oriented identities required by the AFK table hold exactly, rather than only in absolute value.

The signed packet and $\mathbb{Q}(\zeta_{16})$ lie in a convention-selected degree-128 compositum. `scripts/dimension_eight_exact_tcc.gp` verifies trace one, idempotency, and all 784 minors for both shifts.

3.2 Conductor one: quadratic units and exact phases

For $Q = \langle 1,-3,1\rangle$, put

$$C = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A_8 = C^6 = \begin{pmatrix} 377 & -144 \\ 144 & -55 \end{pmatrix}. \tag{18}$$

The one-place and both-place ray groups are C_2^2 and C_2^3. Kopp's exponent is one, and the difference is supported on two quadratic characters. Writing $\phi = (1+\sqrt{5})/2$, their fields and oriented units are obtained from the positive roots

$$h = \sqrt{\phi} > 0, \qquad r = \sqrt{2\phi} > \phi :$$

$$\begin{array}{c|c|c} & \text{absolute equation} & \text{unit} \\ \hline L_0 & h^4 - h^2 - 1 = 0 & u_0 = \phi + h \\ L_1 & r^4 - 2r^2 - 4 = 0 & u_1 = (r+\phi)/(r-\phi). \end{array} \tag{19}$$

Both fields have class number one, with successful unconditional PARI certificates.

After dropping torsion, their fundamental-unit coordinates are

$$\phi = (-2,0), \quad u_0 = (0,-1) \quad \text{in } L_0, \qquad \phi = (1,0), \quad u_1 = (-1,-2) \quad \text{in } L_1.$$

Both determinant indices have absolute value two. This is the exact relative regulator-index calculation; the analytic class-number formula therefore gives

$$L'(0,\chi_0) = \log u_0, \qquad L'(0,\chi_1) = \log u_1. \tag{20}$$

Lemma 5 (Exact magnitude transform)**.** *Let $c = (c_1, c_2)$ be the ray-8 class log of a primitive characteristic. Its differenced partial-zeta derivative is*

$$Z_c'(0) = \frac{1}{2}\left((-1)^{c_2}\log u_0 + (-1)^{c_1+c_2}\log u_1\right). \tag{21}$$

Hence, for

$$x^2 = u_0, \qquad d^2 = \sqrt{u_1/u_0}, \qquad a = xd, \tag{22}$$

the four ray-8 magnitudes are

$$[0,0] \mapsto a, \quad [0,1] \mapsto a^{-1}, \quad [1,0] \mapsto d^{-1}, \quad [1,1] \mapsto d. \tag{23}$$

At modulus four the stabilizer power is two and

$$Z_c'(0) = 2(-1)^c \log u_0, \tag{24}$$

giving magnitudes x^2, x^{-2}; at modulus two the magnitude is one. These cases exhaust all 63 nonzero characteristics.

Proof. For the ray group $G = C_2^2$, Fourier inversion gives

$$Z_c'(0) = \frac{1}{|G|}\sum_{\chi\in\widehat{G}} \overline{\chi(c)}(1-\overline{\chi(R)})L'(0,\chi).$$

The sign class is $R = [0,1]$, so only $\chi_0 = [0,1]$ and $\chi_1 = [1,1]$ survive, each with coefficient $1/2$. Substitution of (20) proves (21) and taking half the exponential gives (23). The identical calculation for the ray-4 group C_2, followed by the exactly audited stabilizer power two, proves (24). The characteristic audit prints 12 occurrences of each ray-8 label, six of each ray-4 label, and three modulus-two labels. $\square$

The continued-fraction word of A_8 is $[3,3,3,3,3,3,0]$. Put $\beta = (3+\sqrt{5})/2$. The exact Euclidean recursion gives the rows

$$\begin{pmatrix} 377 & -144 \\ 144 & -55 \\ 55 & -21 \\ 21 & -8 \\ 8 & -3 \\ 3 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{25}$$

whose values on $(\beta,1)^T$ are

$$\beta^7, \beta^6, \ldots, \beta, 1. \tag{26}$$

Thus all six adjacent period ratios in AFK's continued-fraction cocycle are β. For the characteristic $(p,q)/8$, division of the first period expression by the six successive denominators gives

$$z_i = \frac{q\beta^{i+2} - p\beta^{i+1}}{8}, \qquad 0 \le i < 6, \tag{27}$$

and the outer finite-product index is

$$\frac{-144p + (377-1)q}{8} = -18p + 47q. \tag{28}$$

These identities are checked in exact arithmetic by the archived maximal-tuple audit.

Put $e(t) = e^{2\pi i t}$ and

$$[z;\omega]_n = \begin{cases} \prod_{k=0}^{n-1}(1 - e(z+k\omega)), & n \ge 0, \\ \prod_{k=1}^{-n}(1 - e(z-k\omega))^{-1}, & n < 0. \end{cases} \tag{29}$$

For $s(z) = \lfloor -z \rfloor + 1$, define

$$\Sigma(z) = [z/\beta; -1/\beta]_{-s(z)} e\left(\frac{6(z+s(z))^2 + 6(1-\beta)(z+s(z))}{24\beta}\right) S_2(z + s(z) + 1 \mid \beta, 1). \tag{30}$$

Iterating AFK Definition 1.32 along the rows (25), inserting its outer finite product (28), and replacing each of Kopp's double sines by the reciprocal convention (1) gives

$$\nu_{p,q} = \frac{\tau^{-3(p^2-3pq+q^2)}(-1)^{(1+p)(1+q)}}{[(q\beta - p)/8; \beta]_{-18p+47q}} \prod_{i=0}^{5} \Sigma\left(\frac{q\beta^{i+2} - p\beta^{i+1}}{8}\right), \qquad \tau = -e^{\pi i/8}. \tag{31}$$

For real $t \notin \mathbb{Z}$,

$$1 - e^{2\pi i t} = -2ie^{\pi i t}\sin(\pi t). \tag{32}$$

The fundamental reciprocal double sine is positive on $0 < z < \beta+1$, since it is a ratio of positive Barnes double-gamma values. Equations (2), (31), and (32) therefore reduce every phase to an exact element of $\mathbb{Q}(\beta)/2\mathbb{Z}$. `scripts/dimension_eight_maximal_sign_audit.py` proves for all 63 nonzero characteristics that the coefficient of β cancels, the remaining coefficient of π is integral, and its parity agrees with the signed radical table. Decimal arithmetic is used only to guess a floor; both defining inequalities are then verified exactly in $\mathbb{Q}(\sqrt{5})$. The audit also proves that no finite-product factor or recurrence sine vanishes.

With x, d, a as in Lemma 5, the table uses only

$$\pm 1, \quad \pm x^2, \quad \pm x^{-2}, \quad \pm d, \quad \pm d^{-1}, \quad \pm a, \quad \pm a^{-1}. \tag{33}$$

Exact shared-subfield identities glue it to $\mathbb{Q}(\zeta_{16})$. `scripts/dimension_eight_maximal_exact_tcc.py` verifies, for each shift,

$$\operatorname{Tr}\Pi = 1, \qquad \Pi^2 = \Pi, \qquad 784 \text{ rank-two minors vanish.} \tag{34}$$

3.3 Form-class completion

Solving the admissibility recurrences gives

$$d = 8 \Longrightarrow r = 1, \quad n = 9, \quad K = \mathbb{Q}(\sqrt{5}), \quad j = 2, \quad m = 1, \quad f_2 = 3. \tag{35}$$

Hence the form conductor is exactly $f = 1$ or $f = 3$, and the only discriminants are 5 and 45. The exact wide class calculations are

$$h_{\mathrm{GL}_2}(5) = h_{\mathrm{GL}_2}(45) = 1. \tag{36}$$

For discriminant five, the maximal order also has narrow class number one because ϕ has norm -1. For discriminant 45 only the wide ring class number is used: determinant--1 transformations are allowed, and no narrow-class-number assertion is needed.

If t_M is obtained from t by $M \in \mathrm{GL}_2(\mathbb{Z})$, the unconditional covariance calculation of the companion Paper I [3] sends the complete TCC sum at $(t_M, \boldsymbol{p})$ to that at $(t, \ell M\boldsymbol{p})$, after reindexing its representatives; for $\det M = -1$ both cocycle factors are conjugated. This proves $\mathcal{Z}_t \subseteq \mathcal{Z}_{t_M}$. Applying the same calculation to M^{-1} proves the reverse inclusion. Equations (35)–(36) therefore transport each of the two certified representatives across its own discriminant and complete dimension eight.

Proof of Theorem 2. Section 2 proves both shifts for the discriminant-32 dimension-seven representative and transports them within that discriminant. It explicitly excludes the independent discriminant-8 stratum. Section 3 proves both dimension-eight conductors and transports within each of the only two possible discriminants. □

4 Why dimension six remains different

The dimension-six primitive character has order six. Its projective quotient has order three, rather than the order-two quotient that produces the V_4 CM descent in dimension eight. The alternative index-two induction groups are nonabelian, so no second abelian induction base supplies an imaginary-quadratic elliptic-unit theorem. The finite dimension-six candidate already satisfies all 225 minors; the remaining input is one oriented primitive order-six special value. Thus the gap is analytic, not a finite reconstruction problem.

5 Reproducibility

The final audits were run on Linux 6.8.0-124-generic (x86-64, four virtual AMD EPYC 9354P cores) with PARI/GP 2.15.4, Python 3.12.3, NumPy 1.26.4, python-flint 0.9.0, and FLINT 3.6.0. Representative measurements on that virtual machine were

audit	wall time	peak RSS
dimension-seven closure suite	27.5 s	175 MB
Paper-II archive smoke suite	43.0 s	175 MB.

The submission-specific archive is built deterministically, carries a self-verifying root manifest named `ARCHIVE_CONTENTS.sha256`, and passes its tests after clean extraction. Its Zenodo DOI will be inserted after the immutable deposit; no DOI is claimed in this version.

The supplementary archive contains:

- all dimension-seven conductor-lowering, Shintani-divisor, Arb-height, Artin-label, field-gluing, and exact-minor scripts, including the complete divisor and enclosure transcripts;
- all conductor-three dimension-eight order/maximal-order, linear-reinduction, CM-unit, Arb-orientation, normal-closure, and exact-minor scripts;
- all maximal-order dimension-eight ray, quadratic unit, exact phase, field gluing, and exact minor scripts;

- generated CM-orientation and 63-sign transcripts, the global regression transcript, and SHA-256 checksums; and
- end-to-end regression tests for both dimension-eight strata.

The principal reproduction commands are

```
python3 -m unittest tests.test_dimension_seven_closure
gp -q scripts/dimension_seven_admissible_strata.gp
gp -q scripts/dimension_seven_exact_tcc.gp
python3 scripts/certify_dimension_seven_double_sine.py --tolerance 1e-10
gp -q scripts/dimension_eight_linear_cm_reinduction.gp
gp -q scripts/dimension_eight_cm_unit_lattice.gp
python3 scripts/certify_dimension_eight_cm_orientation.py
gp -q scripts/dimension_eight_cm_real_unit_bridge.gp
gp -q scripts/dimension_eight_exact_tcc.gp
gp -q scripts/dimension_eight_maximal_tuple_audit.gp
gp -q scripts/dimension_eight_maximal_quadratic_units.gp
python3 scripts/dimension_eight_maximal_sign_audit.py
python3 scripts/dimension_eight_maximal_exact_tcc.py
```

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